

Do Stocks Outperform Treasury Bills?

Simulated Multi-Period Returns and Industry Portfolios, after Bessembinder (2018)

Yingni Chang, Yikai Shen

September 2026

Summary

We replicate and extend two tables of Bessembinder (2018). Section 1 revisits the simulation evidence on multi-period returns (Table 1), and Section 2 revisits the bootstrapped portfolios of Table 4 with industry portfolios in place of individual stocks. The main findings are as follows.

- **Daily draws.** With a daily mean of 0.0025%, the median buy-and-hold return is negative and fewer than half of outcomes are positive for every daily $\sigma \geq 0.90\%$, at all horizons. The cause is the low mean, about a tenth of the paper's 0.5% a month, not daily compounding. With the daily mean matched to 0.5% a month, the daily table is close to Table 1.
- **Horizons of 15 and 20 years.** The expected return is the same at every σ , but the median rises with horizon for $\sigma \leq 8\%$ and falls for $\sigma \geq 12\%$. At $\sigma = 20\%$ over 20 years the median is -97.9% and only 11.7% of outcomes are positive, although the expected return is $+231\%$.
- **Low-volatility puzzle.** Compounding explains why volatile assets have worse *typical* (median) long-run outcomes at the same expected return. It does not explain the puzzle itself, which concerns lower *average* returns and alphas.
- **Random industry selection.** A random FF49 industry with equal-weighted stocks beats the VW market in 51.7%, 59.5% and 81.2% of 1-year, 10-year and full-period outcomes, more often than any of the paper's random-stock portfolios. With value-weighted stocks, the shares (49.0%, 46.4%, 39.2%) fall within the paper's range for 25 to 100 stocks.
- **Three-month industry momentum.** Holding the best FF49 industry of the past three months beat the market over 1926–2016 (33,320× vs. 4,715×). But it ranked only at the 63rd percentile of the random paths, suffered a -99.7% drawdown, and had a lower Sharpe ratio (0.40) than the market (0.42).
- **A rule that beats the market.** Hold the **five FF49 industries with the highest return over the previous six months**, equally weighted, value-weighted within each industry, rebalanced monthly. Before costs, it beat the VW market in 8 of 9 decades and in 78.5% of all rolling 10-year windows since 1927 (82.7% after 1995).

All results are before transaction costs. Win rates across historical windows are frequencies on a single history, not probabilities.

1 Simulated multi-period returns (Table 1)

1.1 Method and validation

Returns are i.i.d. $r_t \sim N(\mu, \sigma^2)$, and the T -period buy-and-hold return is $\prod_{t=1}^T (1 + r_t) - 1$. Each experiment simulates 250,000 paths. The 1- and 5-year outcomes come from non-overlapping blocks of the 10-year paths, and the 15- and 20-year outcomes from 240-month paths. We report one set of draws (seed 0); four further independent runs (seeds 1–4) measure Monte Carlo error. Simulated means and SDs are z-tested against exact values from $E[W_T^k]$, where $W_T = \prod_t (1 + r_t)$ (e.g. $E[W_T] = (1 + \mu)^T$). The test is applied only where

the exact skewness is below 10, a screening heuristic for tails light enough for the normal approximation, and every tested cell passes. Draws below -100% are kept. Treating them as an absorbing bankruptcy changes no median, share or percentile. Our monthly replication matches Table 1 to within 0.35 points on the median and 0.18 points on the share positive (Table 2).

1.2 Daily returns

We draw 2,520 daily returns (252 a year) with $\mu = 0.0025\%$ and daily $\sigma = 0.45\%, 0.90\%, \dots, 4.5\%$. We pair these with the paper’s monthly columns $2\%, \dots, 20\%$, since $0.45\% \times \sqrt{21} \approx 2\%$. Table 1 gives the result in the layout of Table 1.

Table 1: Table 1 with **daily** normal returns, $\mu = 0.0025\%$ a day (seed 0, 250,000 paths).

Statistic	Horizon	Daily σ (%)										
		0.00	0.45	0.90	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
		<i>paired monthly column in the paper (%)</i>										
		0	2	4	6	8	10	12	14	16	18	20
Skewness	1y	–	0.22	0.43	0.66	0.90	1.15	1.43	1.74	2.08	2.48	2.94
	5y	–	0.49	1.02	1.65	2.47	3.63	5.37	8.18	12.9	20.8	34.0
	10y	–	0.69	1.51	2.63	4.35	7.21	12.1	20.1	32.4	49.3	70.0
Median (%)	1y	0.6	0.4	–0.4	–1.6	–3.4	–5.6	–8.2	–11.2	–14.5	–18.1	–22.0
	5y	3.2	1.9	–1.9	–7.9	–15.8	–24.9	–34.7	–44.7	–54.3	–63.3	–71.2
	10y	6.5	3.9	–3.7	–15.2	–29.0	–43.6	–57.4	–69.4	–79.1	–86.5	–91.7
% > 0	1y	100.0	52.1	48.9	46.9	45.2	43.6	42.1	40.7	39.2	37.8	36.4
	5y	100.0	54.7	47.6	43.2	39.4	36.0	32.8	29.8	27.0	24.3	21.8
	10y	100.0	56.8	46.7	40.4	35.2	30.6	26.5	22.7	19.2	16.2	13.5
99th pct. (%)	1y	0.6	18.5	38.9	61.9	87.7	116.6	148.7	184.0	222.8	265.0	310.6
	5y	3.2	47.8	106.5	180.9	272.8	382.3	508.0	647.4	795.7	947.2	1,093
	10y	6.5	75.8	175.7	311.1	482.5	684.8	905.4	1,124	1,315	1,452	1,523

Table 2: Comparison with Table 1 at the 10-year horizon. “Matched μ ” is $1.005^{1/21} - 1 = 0.0238\%$ a day, i.e. the paper’s 0.5% a month.

Monthly σ (%)		0	2	4	6	8	10	12	14	16	18	20
Daily σ (%)		0.00	0.45	0.90	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
Median (%)	Paper, monthly	81.9	77.7	65.6	47.3	24.3	0.1	–23.5	–44.6	–62.0	–75.7	–85.3
	Ours, monthly	81.9	77.7	65.5	47.0	24.4	0.1	–23.5	–44.6	–62.0	–75.5	–85.2
	Ours, daily, $\mu = 0.0025\%$	6.5	3.9	–3.7	–15.2	–29.0	–43.6	–57.4	–69.4	–79.1	–86.5	–91.7
	Ours, daily, matched μ	81.9	77.5	64.5	44.9	21.3	–3.5	–27.1	–47.7	–64.3	–76.9	–85.8
% > 0	Paper, monthly	100.0	99.6	87.5	72.1	59.7	50.0	42.1	35.2	29.5	24.2	20.0
	Ours, monthly	100.0	99.6	87.6	72.2	59.9	50.1	42.1	35.2	29.4	24.3	20.0
	Ours, daily, $\mu = 0.0025\%$	100.0	56.8	46.7	40.4	35.2	30.6	26.5	22.7	19.2	16.2	13.5
	Ours, daily, matched μ	100.0	99.4	86.5	70.8	58.5	48.7	40.8	34.1	28.4	23.5	19.3

- **A daily mean of 0.0025% is small.** It compounds to only 0.0525% over 21 days, about a tenth of the paper’s 0.5% a month. Volatility drag therefore wins early. The median is negative and fewer than half of outcomes are positive from daily $\sigma = 0.90\%$ (monthly $\approx 4\%$) at all horizons. In the paper, this happens only from 12% monthly. As a rough guide, the small-return approximation to typical log growth, $\mu - \sigma^2/2$, turns negative near $\sigma \approx 0.71\%$ daily.
- **Daily versus monthly compounding matters little.** With the matched mean, the daily table is close to Table 1: medians are at most 3.7 points below the paper’s. The larger part of the gap is the grid. The daily

σ values correspond to monthly 2.07%–20.9%, not 2%–20%, which lowers the median by up to 5.4 points (five-seed mean). Drawing daily rather than monthly moves the median by at most 0.6 points at 1 year and 2.8 points at 10 years.

- **Skewness.** Right skewness grows with σ and horizon, as in Table 1. At long horizons and high σ , this sample size cannot recover it: with $n = 250,000$ ten-year outcomes, sample skewness cannot exceed $(n - 2)/\sqrt{n - 1} \approx 500$. At 10 years and 20% monthly, the exact value is 662, the paper reports 53.3, and our seeds give 46–186.

1.3 Horizons of 15 and 20 years

We use the paper’s monthly parameters, $\mu = 0.5\%$ and $\sigma = 0\%, 2\%, \dots, 20\%$ (Table 3, Figure 1).

Table 3: Table 1 extended to **15 and 20 years**: monthly normal returns, $\mu = 0.5\%$ (seed 0, 250,000 paths). “Exact” rows are population values from $E[W_T^k]$.

Statistic	Horizon	Monthly σ (%)										
		0	2	4	6	8	10	12	14	16	18	20
Skewness, sample	15y	–	0.83	1.87	3.55	6.77	13.5	27.0	49.6	80.2	115	149
	20y	–	0.97	2.28	4.56	9.04	17.8	33.3	56.6	86.5	121	158
Skewness, exact	15y	–	0.83	1.89	3.64	7.24	16.5	45.6	155	632	3,037	16,689
	20y	–	0.97	2.34	4.98	12.0	36.6	149	793	5,331	43,718	425,770
Median (%)	15y	145.4	136.9	112.9	78.0	38.3	–0.2	–33.4	–59.1	–76.9	–88.1	–94.4
	20y	231.0	215.8	174.0	116.1	54.4	–0.1	–41.9	–69.6	–85.8	–94.2	–97.9
% > 0	15y	100.0	99.9	92.1	76.3	61.8	49.9	40.2	32.0	25.3	19.7	15.1
	20y	100.0	100.0	94.9	79.5	63.6	50.0	38.7	29.6	22.2	16.4	11.7
99th pct. (%)	15y	145.4	340.2	635.1	1,045	1,560	2,145	2,742	3,245	3,540	3,619	3,398
	20y	231.0	546.0	1,046	1,750	2,629	3,562	4,372	4,855	4,887	4,442	3,613
Mean, exact (%)	20y	231.0	231.0	231.0	231.0	231.0	231.0	231.0	231.0	231.0	231.0	231.0

- **The median diverges from the mean with horizon.** The expected 20-year return is 231% at every σ . The median rises with horizon for $\sigma \leq 8\%$ and falls for $\sigma \geq 12\%$. At 10% it stays near zero, and its sign varies across seeds. At 20% it is –97.9% after 20 years, with only 11.7% of outcomes positive.
- **A few extreme paths carry the mean.** The 99th percentile rises with horizon at every σ . Over 20 years it peaks at $\sigma = 16\%$ (4,887%). The exact skewness at 20% is 425,770, but the sample value is 158. Even with 250,000 paths, the seeds’ sample means (192.8%–221.9%) fall short of the true 231%, because the mass sits in paths too rare to sample.

1.4 Comparison with Table 1

The daily and longer-horizon results strengthen the paper’s pattern rather than change it. At a given expected return, higher volatility means a lower median, a lower chance of a gain and a longer right tail, and each effect grows with horizon. This is the mechanism behind the paper’s finding that most stocks underperform T-bills while a few account for the aggregate gains.

1.5 Implications for the low-volatility puzzle

Compounding lowers the typical terminal wealth of volatile assets at equal expected returns, but it does not explain the empirical anomaly, which concerns arithmetic average returns and alphas.

- **What compounding does.** At the same expected 10-year return of 81.9%, the median outcome is +65.5% at $\sigma = 4\%$ but –62.0% at $\sigma = 16\%$, and the probability of a gain falls from 87.6% to 29.4%. Higher

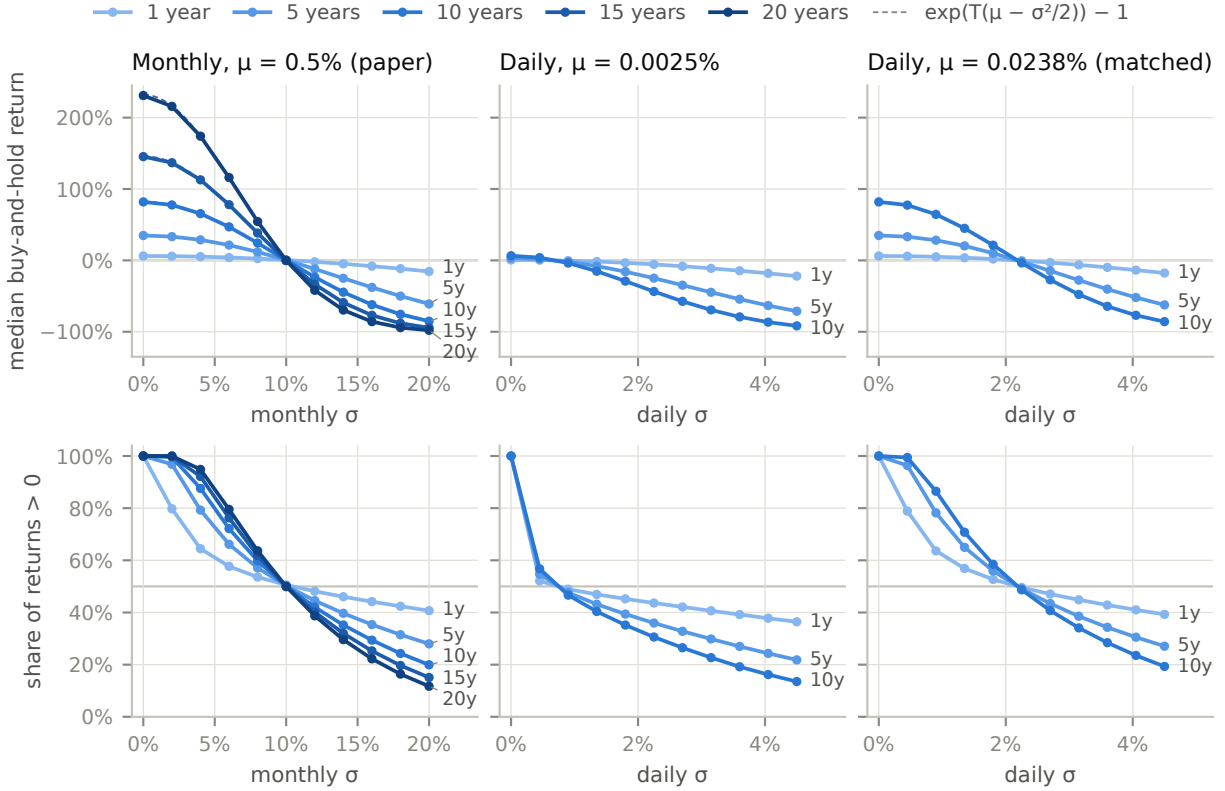


Figure 1: Median buy-and-hold return (top) and share of positive outcomes (bottom) against volatility, by horizon (seed 0). Dashed lines show the lognormal approximation of the median, $e^{T(\mu - \sigma^2/2)} - 1$.

volatility thus lowers typical terminal wealth and the chance of a gain with no difference in expected returns. As a small-return approximation, it lowers typical log growth by about $\sigma^2/2$ per period (under the literal normal model, returns below -100% have positive probability, so $E[\log(1 + r)]$ is not defined). These are comparisons of marginal distributions; we do not estimate how often a high-volatility asset trails a low-volatility one.

- **What it does not explain.** The anomaly (Ang et al. 2006; Baker, Bradley and Wurgler 2011; Frazzini and Pedersen 2014) is that high-volatility or high-beta stocks earn lower *average* returns and negative alphas. The simulation holds the arithmetic mean fixed by construction, so it cannot produce that.
- **An indirect link.** Compounding makes high-volatility stocks strongly right-skewed, like lotteries. If investors prefer positive skewness, they overpay for such stocks and lower their expected returns. This is consistent with Table 1 but not tested here. Two further caveats apply: the simulation uses total volatility, whereas the literature sorts on idiosyncratic volatility or beta; and cumulative-wealth comparisons alone cannot identify differences in expected returns.

2 Industry portfolios (Table 4)

2.1 Data

We use Ken French’s 49 (main) and 12 (robustness) industry portfolios, with monthly equal-weighted (EW) and value-weighted (VW) returns, from the August 2026 CRSP release. The VW market return is $\text{Mkt} - \text{RF} + \text{RF}$, and the T-bill return is RF . The sample is the paper’s, July 1926 to December 2016 (1,086 months). French assigns stocks to industries each June but does not date his firm counts, so we assume that a positive July firm count means the industry had members at the preceding June formation, and treat it as eligible for

that July–June year. This gives 40 FF49 industries in 1926 and all 49 from 1969; FF12 always has 12. The universe is thus reconstructed from revised data; it does not show what an investor could have known at the time.

2.2 Evaluation method

The evaluation follows the paper. The horizons are the 90 calendar years 1927–2016, the paper’s nine decades (the first running from July 1926 to December 1936) and the full period. A strategy “beats” a benchmark when its holding-period wealth is strictly higher over the same window. For random strategies, statistics are pooled over all path $\times$ window outcomes. The paper does not document how it pools windows, so this is our choice. For a deterministic strategy they are frequencies across historical windows. Monte Carlo standard errors of the pooled shares are at most 0.35 points.

2.3 Random industry selection

Each month we draw one eligible industry uniformly at random, hold it for the month, then draw again. We run 20,000 paths with seed 0. The same draws are applied to EW and VW returns, so the two differ only in how stocks are weighted within the industry. Conditional on history, the expected wealth of this strategy is exact:

$$E[W] = \prod_t \frac{1}{|A_t|} \sum_{i \in A_t} (1 + r_{i,t}), \quad (1)$$

where A_t is the eligible set. This is the wealth of an equal-weight portfolio of industries, rebalanced monthly. Every formally tested simulated mean passes a z-test against it (a normal approximation at a nominal 0.1% Bonferroni family-wise level per variant, with a minimum threshold of 4). The heavy-tailed FF49 full-period rows are reported as diagnostics only.

Table 4: Table 4 with **random industries** in place of random stocks (20,000 paths, July 1926 to December 2016). “Ours” is pooled across paths and windows. The paper’s columns are its 1-, 5-, 25-, 50- and 100-stock portfolios. Full-period mean and median are in whole percent. Skewness values are Monte Carlo estimates; the FF49 full-period values are especially sensitive to rare paths (exact values 80.3 EW and 78.6 VW).

Horizon	Statistic	Random industry (ours)				Random stocks (paper)				
		FF49 EW	FF49 VW	FF12 EW	FF12 VW	1	5	25	50	100
1 year	Mean (%)	17.11	13.25	16.76	12.53	16.56	13.16	12.26	12.08	11.95
	Median (%)	13.99	12.26	16.04	13.17	4.06	10.72	12.52	12.90	13.18
	Skewness	1.75	1.32	0.49	0.06	6.99	1.08	0.10	−0.09	−0.21
	% > 0	67.75	68.67	69.38	72.36	53.59	64.33	70.00	71.21	72.00
	% > T-bill	63.59	63.64	65.68	66.76	50.79	59.98	64.94	66.19	67.09
	% > VW market	51.69	48.97	54.47	50.42	42.86	47.20	48.69	49.10	49.28
10 years	Mean (%)	268.59	196.98	262.29	191.73	245.38	191.80	181.88	179.80	178.05
	Median (%)	203.40	160.05	227.28	166.81	27.72	123.64	139.77	140.09	137.60
	Skewness	3.71	2.77	1.56	0.99	65.03	9.03	1.64	1.15	0.90
	% > 0	95.77	90.97	99.32	98.16	56.18	83.60	95.96	98.38	99.57
	% > T-bill	90.34	82.57	97.72	91.80	47.77	72.29	86.86	90.70	93.08
	% > VW market	59.54	46.39	69.50	51.81	29.38	40.77	45.37	46.70	47.54
Full period	Mean (%)	7,573,243	1,173,532	6,854,963	994,527	949,826	895,497	635,547	586,071	544,181
	Median (%)	2,051,121	317,262	4,041,583	596,943	9.5	94,936	317,456	384,332	421,749
	Skewness	23.32	15.82	5.75	4.91	96.45	47.24	10.02	4.40	2.95
	% > 0	100.00	100.00	100.00	100.00	50.76	99.94	100.00	100.00	100.00
	% > T-bill	100.00	99.89	100.00	100.00	27.45	96.48	100.00	100.00	100.00
	% > VW market	81.23	39.22	97.97	56.75	3.97	22.68	36.81	40.94	43.29

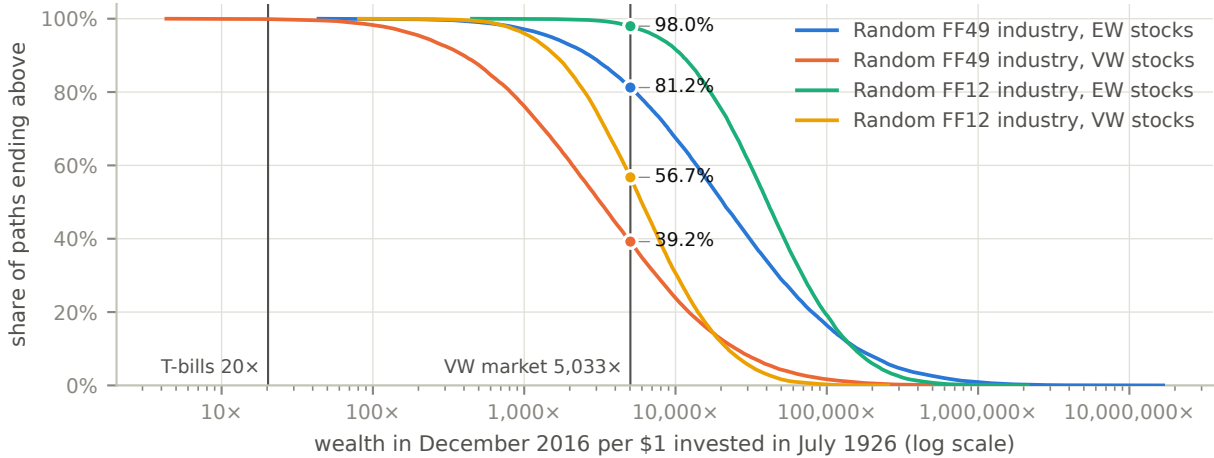


Figure 2: Full-period wealth of the 20,000 random-industry paths. Each curve gives the share of paths ending above a given wealth level. Dots mark the share that beat the VW market.

- **Equal-weighted industries beat the market more often than any stock portfolio in the paper.** FF49 EW beats the market in 51.7%, 59.5% and 81.2% of outcomes, against at most 49.3%, 47.5% and 43.3% for the paper’s 100-stock portfolio. It is far from the single stock’s 3.97% over the full period. It also beats T-bills in all 20,000 full-period paths, where only 27.5% of the paper’s single stocks did.
- **With value weighting, market-beating shares resemble those of diversified stock portfolios.** FF49 VW’s shares (49.0%, 46.4%, 39.2%) lie inside the paper’s 25- to 100-stock range at every horizon. This concerns that one statistic, not the whole distribution. Its 1-year skewness is 1.32, against 6.99 for a single stock, which is consistent with diversification within industries. The comparison does not isolate that effect, however, and the data vintage and construction also differ from the paper’s.
- **Equal weighting within the industry adds 2.7, 13.2 and 42.0 points** to the share beating the market (EW vs. VW, same draws). Its exact 1-year mean is 17.1% against 13.2% for VW, before the unmeasured costs of maintaining equal weights. The broader FF12 industries have higher shares and much less skewness than FF49.

2.4 Three-month industry momentum

Each month t we hold the industry with the highest compounded return over months $t - 3$ to $t - 1$, among industries that are eligible in month t and have all three lookback returns. Ties go to the first in column order. Ranking and holding use the same returns (EW or VW), and the first holding month is October 1926. Momentum is a single path, so we compare it with the 20,000 random paths of Section 2.3 over the same months (Tables 5 and 6, Figure 3). The lookback requirement makes momentum’s candidate set smaller than the random baseline’s in 21 FF49 months, just after an industry enters; restricting the baseline accordingly would raise its exact mean wealth by under 1%.

- **FF49 EW momentum beat the market and the median random path, but not the mean of random selection.** It grew \$1 to 33,320 $\times$, against 4,715 $\times$ for the market and 19,430 $\times$ for the median random path, which places it at the 63rd percentile. It fell short of the 69,258 $\times$ expected wealth of random selection (Eq. 1). It beat the market in 54.4% of years, 6 of 9 decades and 69.6% of rolling 10-year windows.
- **The path was extreme.** Wealth fell 99.7% from 1929 to 1942, and volatility was 35.9%. The Sharpe ratio of 0.40 is below the market’s 0.42 and the equal-weight portfolio’s 0.48. Momentum held an industry with at most five firms in 34% of months, against 15% under random selection, and its most frequent holdings were Gold, Coal and Smoke. A plausible but untested mechanism is that ranking on noisy three-month

Table 5: Three-month industry momentum against random industry selection, October 1926 to December 2016. The random mean is exact (Eq. 1). The percentile is the share of random paths with lower full-period wealth. Window columns give the share of calendar years, paper decades and rolling 10-year windows in which the strategy beat the VW market, shown as momentum/random.

Variant	Wealth per \$1			Percentile [95% CI]	% of windows > VW market		
	Momentum	Random median	Random mean		Years	Decades	Rolling 10y
FF49 EW	33,320	19,430	69,258	63.16 [62.49, 63.83]	54.4 / 51.7	66.7 / 59.6	69.6 / 61.7
FF49 VW	113	2,968	10,993	2.22 [2.02, 2.43]	47.8 / 49.0	33.3 / 46.4	46.5 / 48.3
FF12 EW	39,337,330	38,766	65,110	100.00 [99.98, 100.00]	71.1 / 54.5	100.0 / 69.7	94.6 / 70.8
FF12 VW	136,822	5,574	9,237	99.94 [99.90, 99.97]	66.7 / 50.4	88.9 / 51.8	83.9 / 51.8
VW market	4,715						
T-bills	20						

Table 6: Performance, October 1926 to December 2016, before costs. Return and volatility are annualized (%). The Sharpe ratio is over T-bills. Turnover is one-way turnover across industry portfolios (% a year). “Equal weight” holds every eligible industry equally, which is the expected outcome of random selection.

Strategy	Wealth	Return	Volatility	Sharpe	Max drawdown (peak to trough)	Turnover
Momentum, FF49 EW	33,320	12.2	35.9	0.40	−99.7 (1929-07 to 1942-03)	761
Momentum, FF49 VW	113	5.4	32.9	0.23	−99.2 (1929-08 to 1941-04)	811
Momentum, FF12 EW	39,337,330	21.4	26.4	0.74	−83.7 (1929-09 to 1932-05)	574
Momentum, FF12 VW	136,822	14.0	20.6	0.58	−71.1 (1929-09 to 1932-05)	677
Equal weight, FF49 EW	69,258	13.1	24.9	0.48	−85.0 (1929-08 to 1932-05)	19
Equal weight, FF49 VW	10,993	10.9	20.7	0.44	−85.3 (1929-08 to 1932-06)	20
Equal weight, FF12 EW	65,110	13.1	24.0	0.49	−84.7 (1929-08 to 1932-05)	13
Equal weight, FF12 VW	9,237	10.6	18.5	0.46	−83.4 (1929-08 to 1932-05)	13
VW market	4,715	9.8	18.6	0.42	−83.7 (1929-08 to 1932-06)	–
T-bills	20.14	3.4	0.9	–	−0.1 (1938-10 to 1941-04)	–

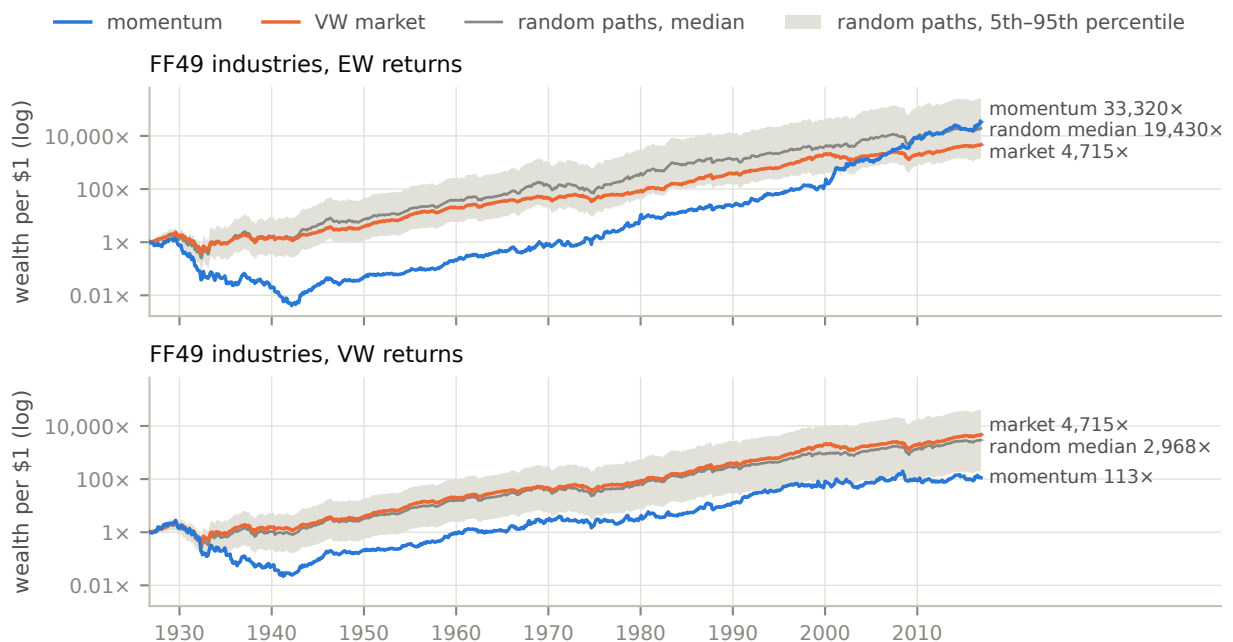


Figure 3: Wealth of FF49 three-month momentum, the VW market and the random-industry paths (median and 5th–95th percentile band), October 1926 to December 2016.

returns favors the most volatile industries, which in FF49 often have few firms.

- **The results depend on the universe and the weighting.** FF49 VW momentum ended at 113 $\times$ (2nd percentile). EW and VW momentum rank on different returns and hold different industries in 696 of 1,083 months, so this gap reflects both rankings and holding returns, not weighting alone. With the broad FF12 industries, momentum beat every random path with EW returns (Sharpe 0.74) and 99.9% of them with VW returns. Turnover is 574%–811% a year, so trading costs would matter.

2.5 A rule that beats the value-weighted market over ten years

We seek a rule with at least an even chance of beating the VW market over ten years, before transaction costs. Two candidate rules and a pass/fail criterion were specified ex ante, before the analyses of Sections 2.3 and 2.4, and were not tuned.

- **(a) Equal weight:** hold every eligible FF49 industry equally, value-weighted within each industry, rebalanced monthly. This is the expected outcome of random FF49 VW selection.
- **(b) Top 5, 6-month momentum:** each month, hold the five FF49 industries with the highest return over months $t - 6$ to $t - 1$ (Moskowitz and Grinblatt’s industry momentum), equally weighted and value-weighted within each industry. Holding five industries rather than one diversifies away the single-industry risk documented in Section 2.4, and value weighting limits the influence of tiny firms.

The rules are evaluated from January 1927 to August 2026. **Criterion:** a rule passes if it beats the VW market in at least 50% of the paper’s nine decades *and* of the 249 rolling 10-year windows that lie entirely within January 1996 to August 2026. Because Sections 2.3 and 2.4 use data through 2016, this holdout is a retrospective validation. Adjacent rolling windows share 119 months, and the holdout contains only three non-overlapping decades, so its 249 windows are not independent confirmations. January 2017 to August 2026 (116 months, with no complete 10-year window) lies outside the earlier samples. It is reported separately as a post-2016 evaluation of the prespecified candidates, not as an independent test of the final choice: those months also enter the screen and the comparison between the two rules.

Table 7: Win rates against the VW market over 10-year windows (strictly higher holding return), before costs. The two rows marked “screen” form the pass/fail criterion.

Measure	(a) equal weight			(b) top 5, 6-month	
	Windows	Wins	Win rate	Wins	Win rate
Paper decades to 2016 (screen)	9	7	77.8%	8	88.9%
Holdout rolling 10-year windows (screen)	249	154	61.8%	206	82.7%
Pre-1996: paper decades ending by Dec 1986	6	5	83.3%	6	100.0%
Pre-1996: rolling windows ending by Dec 1995	709	549	77.4%	569	80.3%
Holdout decades 1996–2005, 2006–2015, 2016–2025	3	2	66.7%	3	100.0%
All rolling 10-year windows	1,077	748	69.5%	845	78.5%

Recommended rule: (b). Both rules pass the criterion (Table 7), but (b) wins at least as often as (a) on every measure:

- 8 of 9 decades, 206 of 249 holdout windows, and 845 of all 1,077 rolling 10-year windows since 1927 (78.5%);
- 14.4% a year against the market’s 10.3% over 1927–2026, with a Sharpe ratio of 0.58 against 0.45;
- a win in the post-2016 period (4.61 $\times$ vs. 3.92 $\times$), although with a lower Sharpe ratio (0.72 vs. 0.81).

Rule (a), by contrast, has lost every rolling window ending since March 2019. Two caveats apply. Rule (b)’s 78.5% is a historical frequency on overlapping windows of one history, not an established probability that it

Table 8: Performance by period, before costs. Wealth is per \$1. Return and volatility are annualized (%). Turnover is one-way across industries (% a year). Jan 2017–Aug 2026 is the post-2016 evaluation period (116 months).

Period	Strategy	Wealth	Return	Volatility	Sharpe	Max drawdown	Turnover
Jan 1927–Aug 2026	(a) equal weight	35,774	11.1	20.3	0.46	−85.3	20
	(b) top 5	662,986	14.4	22.0	0.58	−82.6	455
	VW market	17,985	10.3	18.4	0.45	−83.7	–
Jan 1927–Dec 1995	(a) equal weight	1,458	11.1	21.9	0.42	−85.3	20
	(b) top 5	9,548	14.2	22.6	0.54	−82.6	446
	VW market	816	10.2	19.4	0.41	−83.7	–
Jan 1996–Aug 2026	(a) equal weight	24.53	11.0	16.4	0.58	−52.6	21
	(b) top 5	69.44	14.8	20.5	0.67	−50.4	476
	VW market	22.04	10.6	15.7	0.58	−50.3	–
Jan 2017–Aug 2026	(a) equal weight	3.32	13.2	17.1	0.68	−25.9	21
	(b) top 5	4.61	17.1	22.2	0.72	−23.1	471
	VW market	3.92	15.2	16.1	0.81	−24.8	–

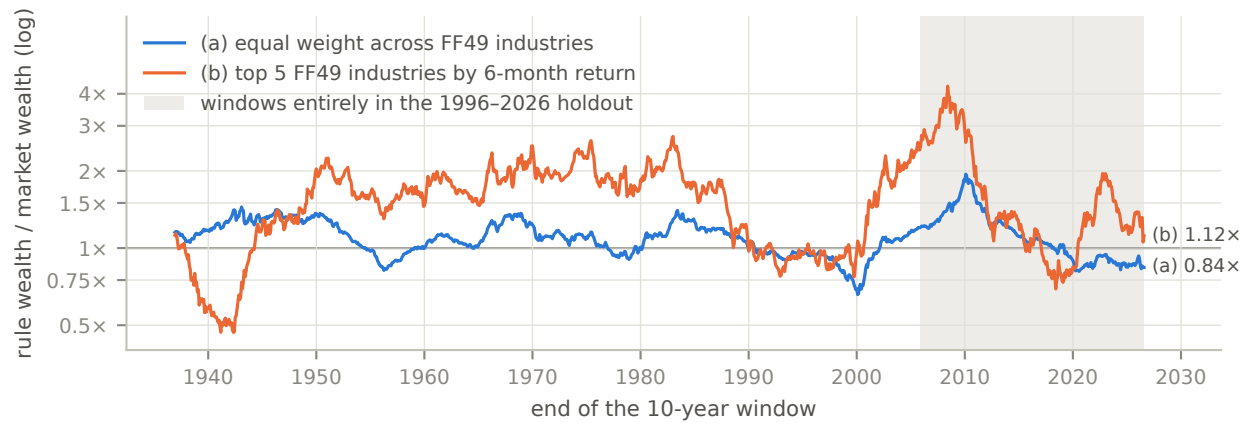


Figure 4: Each rule’s rolling 10-year wealth relative to the VW market, at the window’s end; above 1×, the rule won.

beats the market over the next decade. And its turnover of about 455% a year is 23 times rule (a)’s, so trading costs, which are not modeled here, would reduce its edge.

3 Limitations

All results are before transaction costs, and turnover counts trades between industries only. The returns come from the current, revised CRSP data (CIZ format), not the FIZ-era data the paper used, and the size of that difference is not measured. Industries are not stocks, and FF12 is not a GICS sector scheme. Pooling across windows and dropping the partial year 1926 are our choices; the paper documents neither. The industry simulations use a single seed.

Code and data. All computations are in `hw2_analysis.ipynb`, using the tested modules in `src/.report/make_tables.py` and `report/make_figures.py` rebuild these tables and figures from `output/` and check them against it.

References. Ang, A., R. Hodrick, Y. Xing and X. Zhang (2006), The cross-section of volatility and expected returns, *Journal of Finance* 61. Baker, M., B. Bradley and J. Wurgler (2011), Benchmarks as limits to arbitrage: understanding the low-volatility anomaly, *Financial Analysts Journal* 67. Bessembinder, H. (2018), Do stocks outperform Treasury bills?, *Journal of Financial Economics* 129. Frazzini, A. and L. Pedersen (2014), Betting against beta, *Journal of Financial Economics* 111. Moskowitz, T. and M. Grinblatt (1999), Do industries explain momentum?, *Journal of Finance* 54.